

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Outperform Optical and ImageNet Transfer Across ViT and CNN?

John Roth

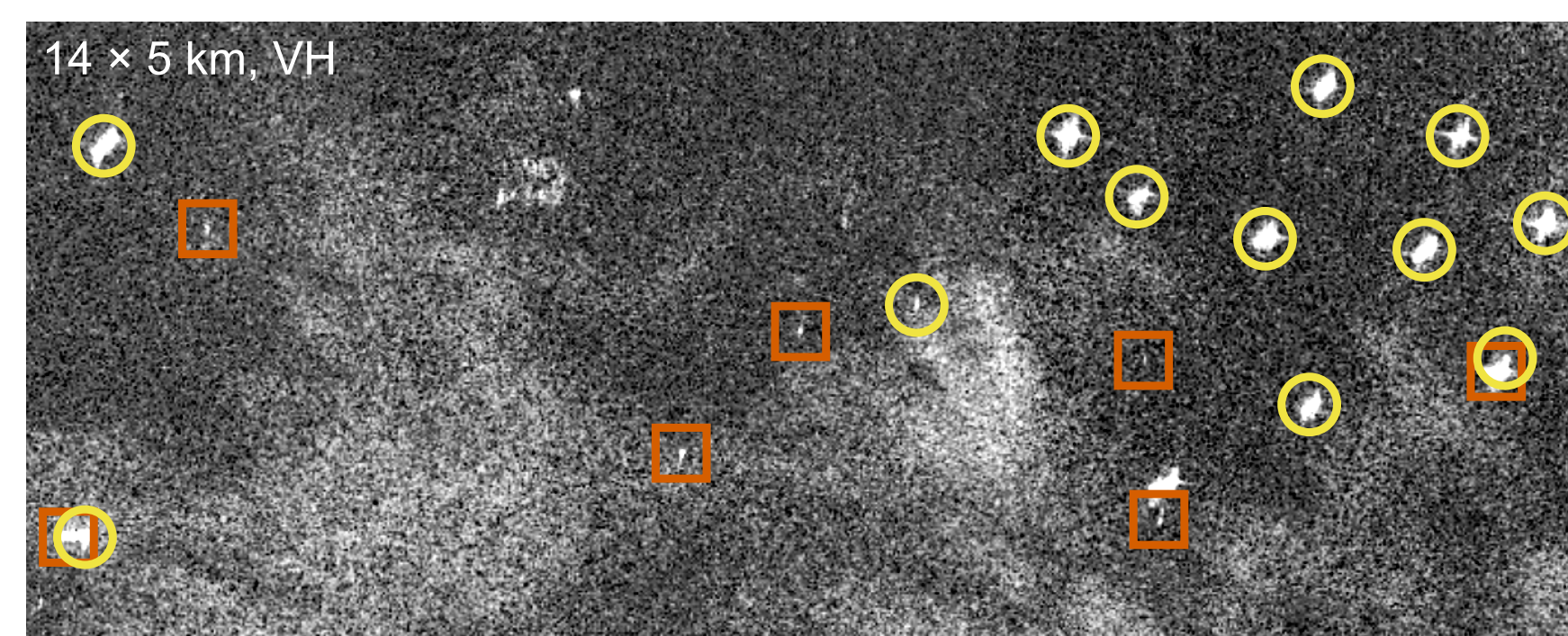
Kyle Wagner

Liv d'Aliberti

Johns Hopkins University

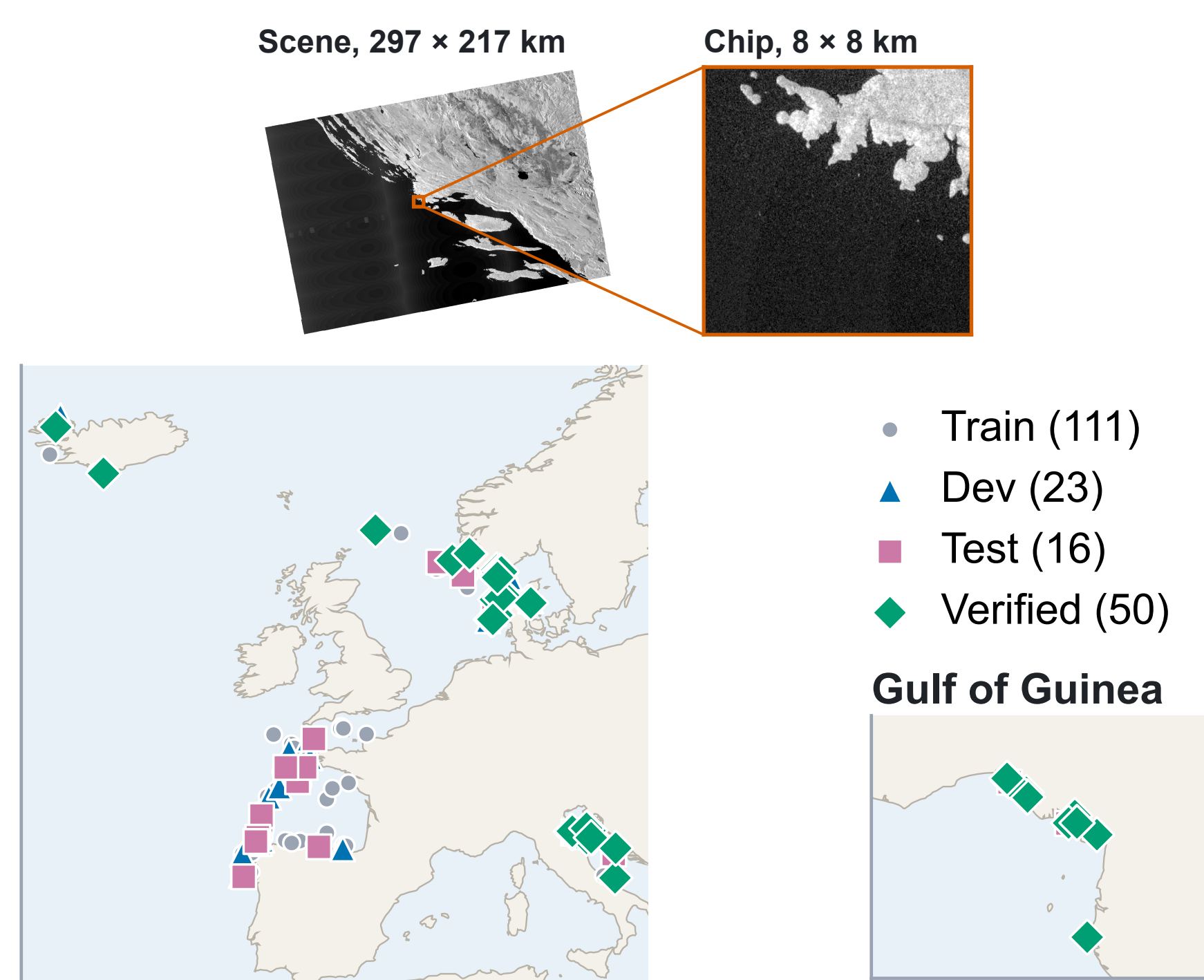
Answer: no. SAR pretraining beats optical in 6 of 16 comparisons and ImageNet in 3 of 16 (ImageNet wins 7); its clearest edge is the CNN with 12–28 labeled scenes.

Motivation



Dark vessels send no AIS, yet radar [2] still sees them. Human-checked labels are scarce: which pretraining stretches them?

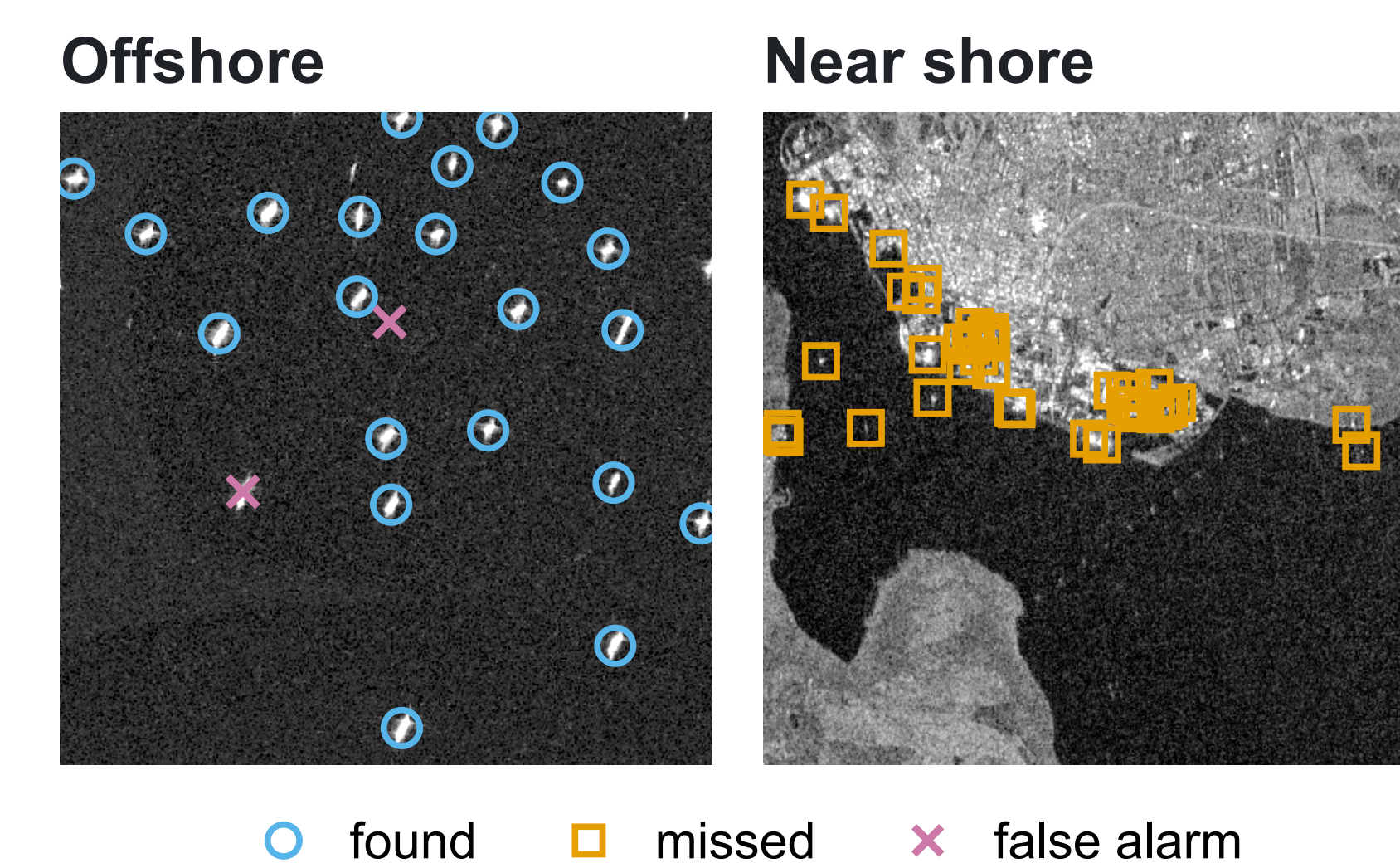
Data: xView3-SAR [1]



■ **Machine-made labels** (AIS + automated detection) for train, dev and test; they miss vessels.

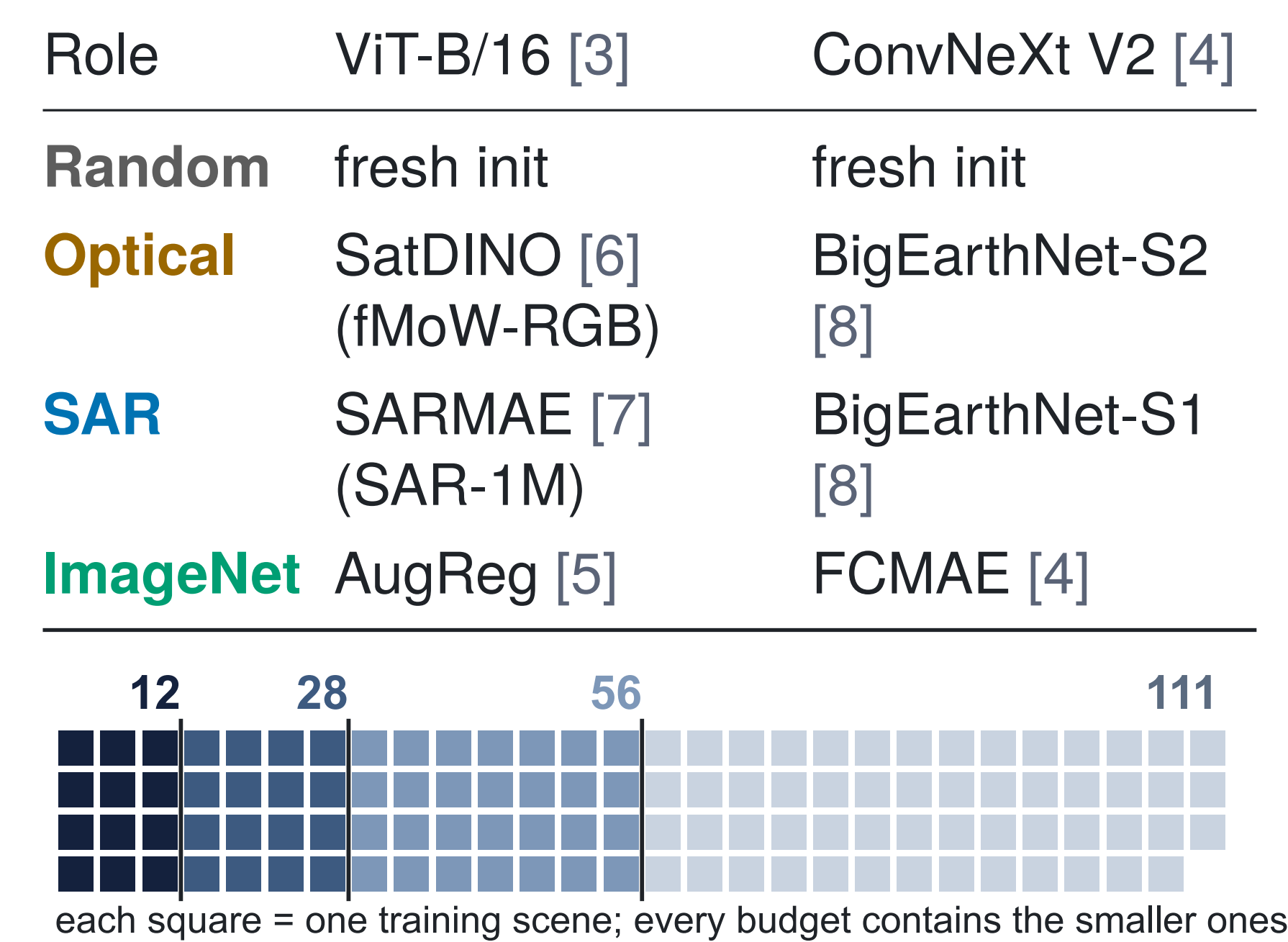
■ **50 human-verified scenes:** 8,642 vessels.

What the best detector finds

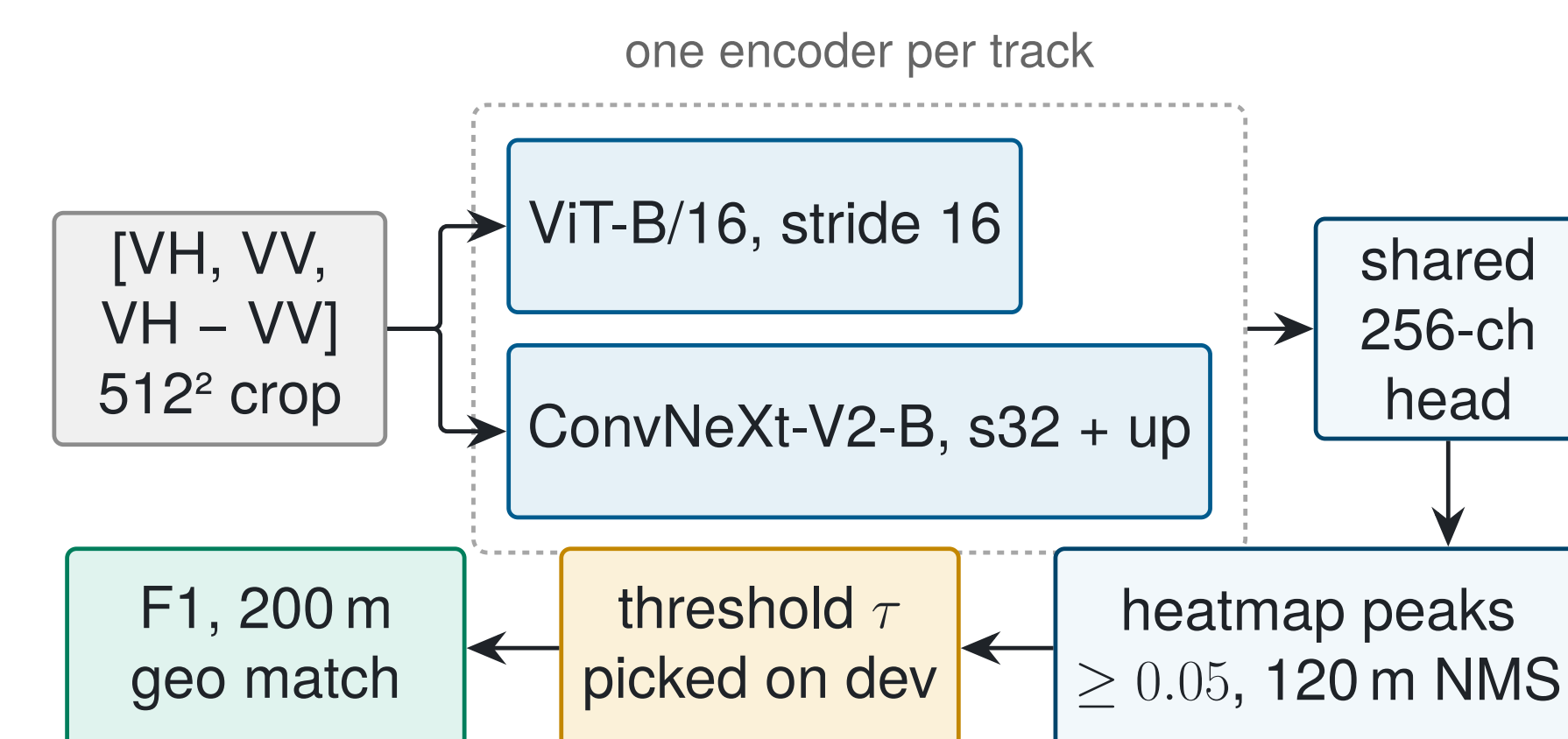


Best cell (ImageNet CNN, 111 scenes), 8 × 8 km verified crops. Verified F1: 0.724 offshore, 0.026 within 2 km of shore.

8 arms × 4 budgets = 32 cells



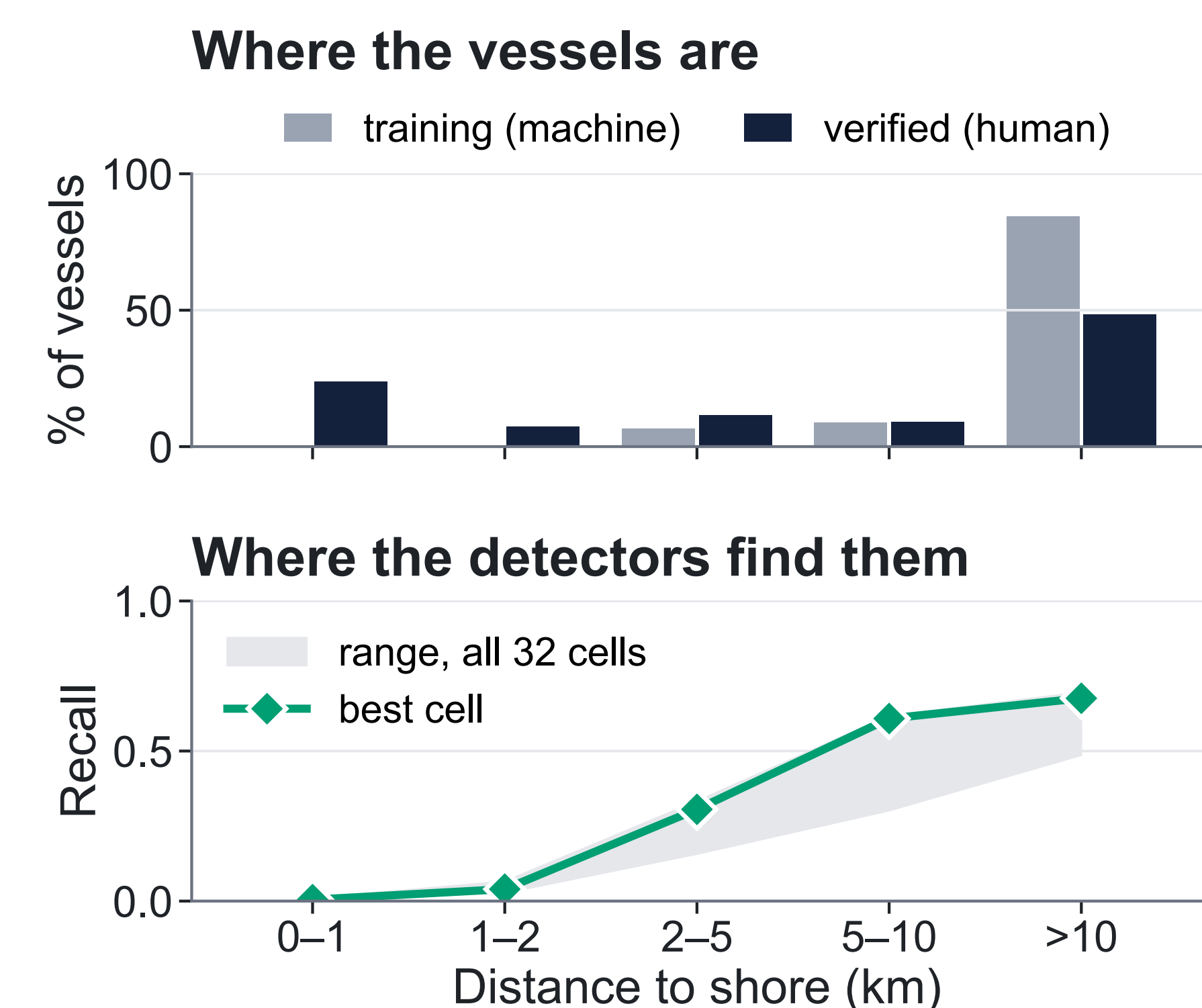
One shared point detector



■ CenterNet-style heatmap [9], focal loss [10]; one recipe and seed for every cell.

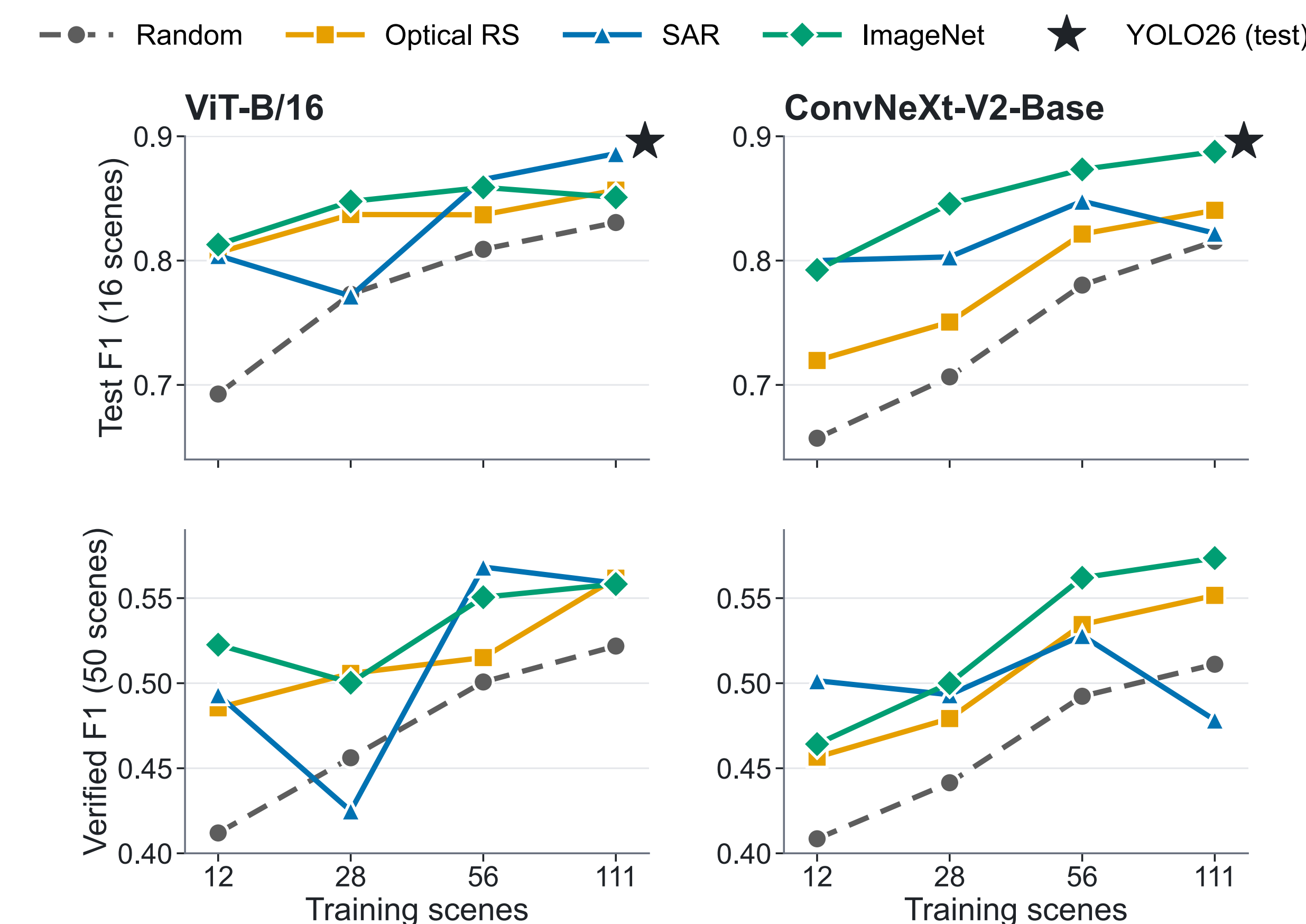
■ 8 dev scenes pick checkpoint and threshold; test and verified scenes are scored **once**.

Recall falls toward the coast



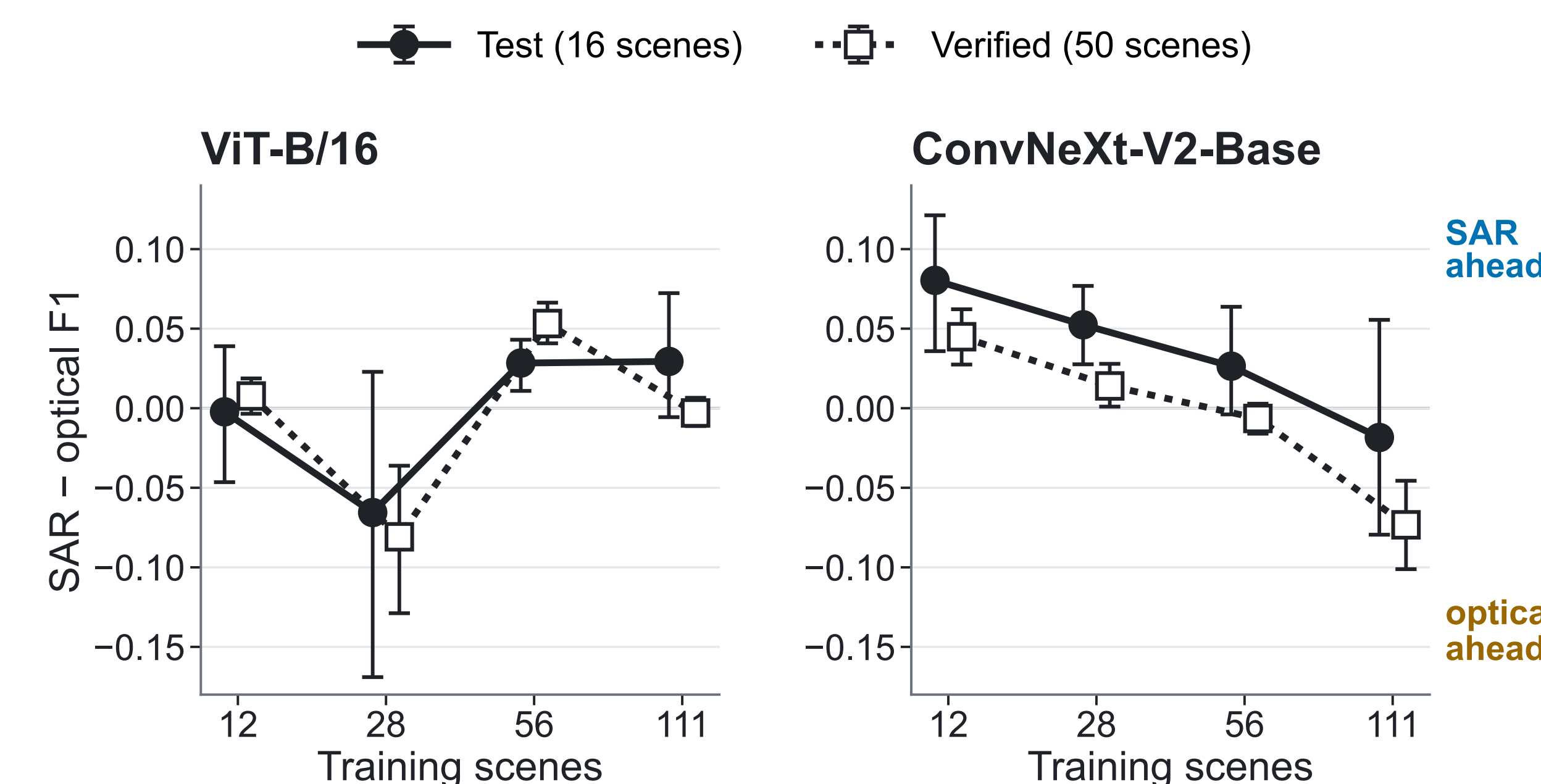
50 verified scenes; the band spans all 32 cells. Best-cell recall: 0.676 beyond 10 km, 0.005 within 1 km.

Pretraining helps most when labels are scarce



F1 with 200 m matching, each cell scored once at the threshold it picked on 8 dev scenes. Top: 16 test scenes; bottom: 50 verified scenes. ★ YOLO26 at 111 scenes, test only (0.896). Zero-shot LocateAnything-3B: 0.122 F1 on dev chips.

SAR vs. optical: no stable winner



How to read: each point is SAR F1 minus optical F1; above 0, SAR wins, and a bar clear of 0 is a win. Solid: test; dotted: verified. Bars: 95% intervals over 10,000 scene resamples; retraining moved test F1 by up to 0.043.

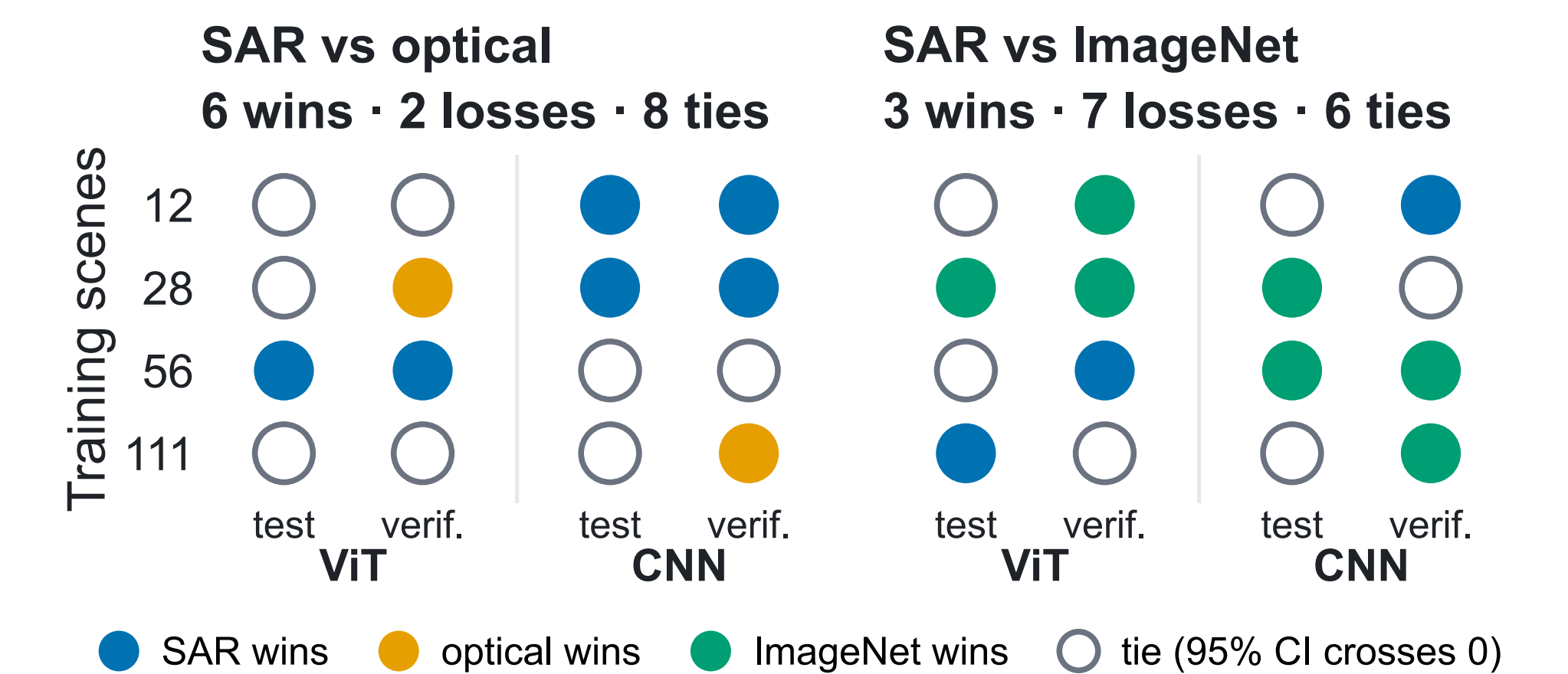
References

- [1] F. S. Paolo et al. xView3-SAR: Detecting dark fishing activity using SAR imagery. NeurIPS 2022.
- [2] R. Torres et al. GMES Sentinel-1 mission. Remote Sens. Environ. 2012.
- [3] A. Dosovitskiy et al. An image is worth 16×16 words. ICLR 2021.
- [4] S. Woo et al. ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. CVPR 2023.
- [5] A. Steiner et al. How to train your ViT? TMLR 2022.
- [6] J. Straka and I. Gruber. SatDINO: A deep dive into self-supervised pretraining for remote sensing. arXiv:2508.21402, 2025.
- [7] D. Liu et al. SARMAE: Masked autoencoder for SAR representation learning. CVPR 2026.
- [8] K. N. Clasen et al. reBEN: Refined BigEarthNet dataset. IGARSS 2025.
- [9] X. Zhou, D. Wang, and P. Krähenbühl. Objects as points. arXiv:1904.07850, 2019.
- [10] T.-Y. Lin et al. Focal loss for dense object detection. ICCV 2017.

Findings

1 Pretraining pays off with few labels: +0.063 to +0.143 test F1 at 12 scenes.

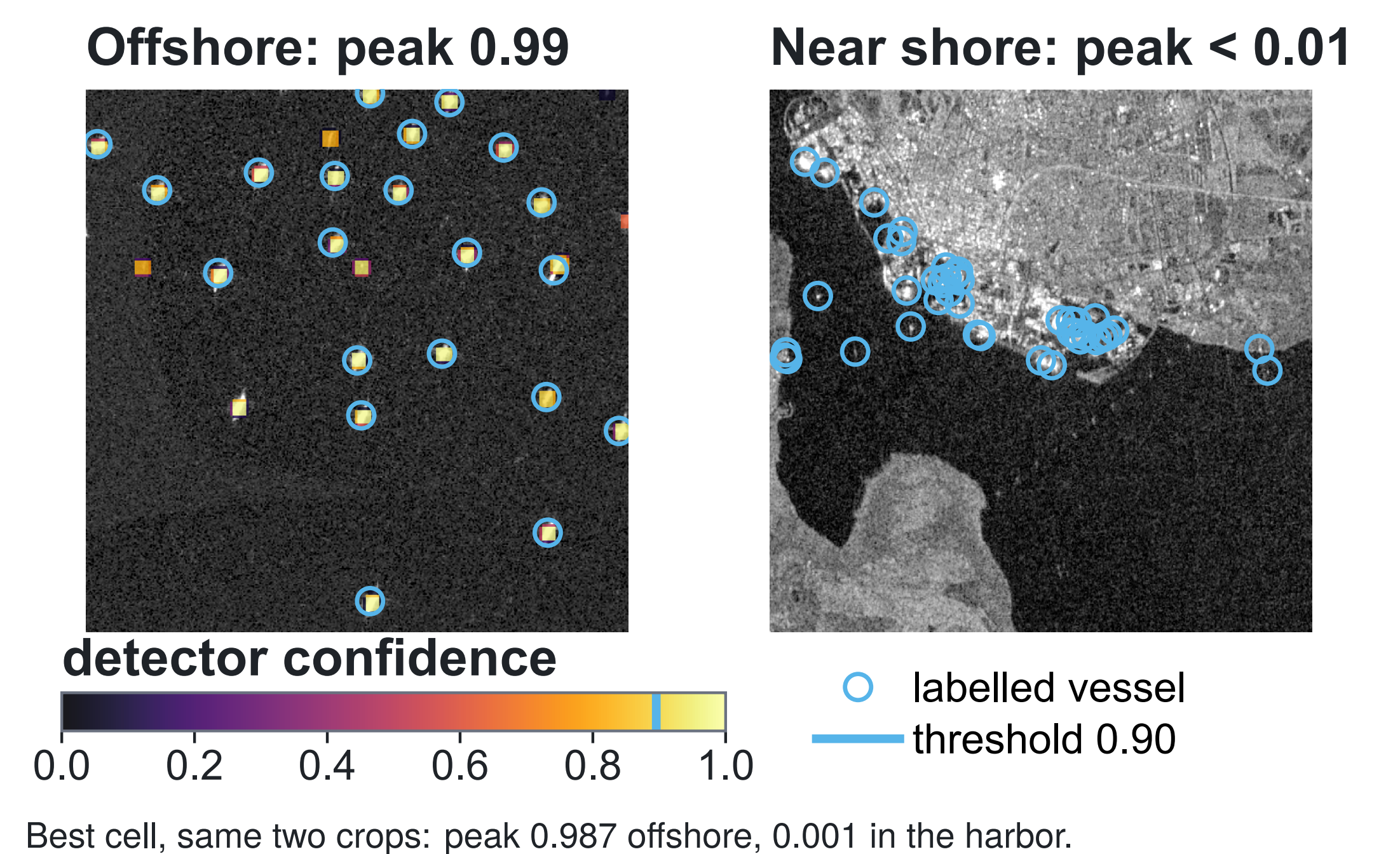
2 SAR pretraining rarely wins:



3 The input adapter matters: a seeded 3-channel stem lifts BigEarthNet-S2 above random at every budget.

4 No detector works near shore: recall ≤ 0.021 within 2 km in every cell.

Inside the detector



Best cell, same two crops: peak 0.987 offshore, 0.001 in the harbor.

Limits

- One seed per cell; 8 dev scenes for selection.
- Machine-made test labels; in-region test scenes.
- Checkpoints also differ in objective and data.

Open questions & next steps

- Would joint SAR + optical foundation models win?
- Does SAR trail ImageNet with matched pretraining?
- **Next:** near-shore labels, shoreline input, seeds.